

Aws Route 53

You can use amazon Route 53 to register new domain transfer existing domain route traffic for your domain to your AWS and external resources and monitor the health of your resources

Route 53 functions

1. DNS management
2. Traffic management
3. Availability Monitoring
4. Domain Registration

Route- 53 Perform three main functions

1. Register a domain
 2. As a DNS it routes internet traffic to resources for your domain
 3. Check the health of your resource, Route- 53 sends automated request over the internet to a resource (can be web server) to verify that the server is reachable functional or available
- Also,
- you can choose to receive notification when a resource becomes unavailable and choose And choose to route internet traffic and from unhealthy resources
 - You can use Route 53 for any combination of these functions
 - For e.g. You can use Route 53 both to register your domain name and to route internet traffic for the domain
 - You can use Route 53 to route internet traffic for a domain that you register with another domain register

When you register a domain with route 53 the service automatically makes itself the DNS services for the domain by doing the following

- It creates a hosted zone that has the same as your domain
- It assigns a set of four name server to the hosted zone unique to the account
 - When someone use a browser to access your website this name server information the browser where to find your resources such as a web server or an amazon s3 bucket
 - It gets the name server from the hosted zone and adds them to the domain
- Aws Support
 1. Generic Top-Level Domain
 2. Geographical Top-Level Domain

Register a Domain with Route 53

- You can register a domain with route 53 if the TLD is included on the supported TLD list
 - If the TLD is not included you cannot register the domain with the route 53
 - You can use Route 53 as the DNS services for any domain even if the TLD for the domain is not included on the supported TLD list
- Note:
- Each Amazon Route 53 account is limited to a maximum of 500 hosted zones and 10000 resources record sets per hosted zone. You can increase this limit by requesting to Aws

Delegate to Route 53

- This step connects everything and make it works
- **Connect the domain name to the route 53 hosted zone - this is called delegation**
 - Update your domain register with the correct name server for your route 53 hosted zone
 - No other customer hosted Zone with share this delegation set with you
 - Doing this means route 53 DNS services will be serving DNS traffic for domain of the hosted zone
 - If you registered your domain with a diff register you need to configure the route53 list in your register DNS database for your Domain
- **If you are using another domain provider and you did all the changes**
 - When you migrate from one DNS provider to another for an existing domain this change can take up to 48h to be effective
 - This is because name server DNS records are typically cached across the DNS system globally on the internet for up to 48 h
- **Transferring a domain to route 53**
 - You can transfer a domain to route53 if the TLD is included on the following list
 - If the TLD is not included, you cannot transfer the domain to route 53
 - For most TLD you need to get an authorization code from the current registrar to transfer a domain

Route 53 Hosted Zone

- A route 53 hosted zone is a collection of records for a specified domain
- You create a hosted zone or a domain and then you create tell domain name system how you want traffic to be routed for that domain
- Basically, a hosted zone is a container that holds information about how you want to route traffic for a domain and its subdomain
- You can create public (internet) hosted zone or private (internal DNS) hosted zone
- For Each public hosted zone that you create amazon route 53 automatically create a name server (ns) record and a start of authority (SOA) record
- Route 53 automatically creates a name server (ns) record with the same name as your hosted zone
- It lists the four name servers that are the automatically name servers for your hosted zone
- Do not add change or delete name server in this record
- When you create a hosted zone , Amazon rout 53 automatically creates name server(ns) record and a start of authority (soa) record for the zone
- The NS record identifies the four name server (ns) that you give to your register or your DNS services so that DNS queries are routed to route 53 name servers
- By default Rout 53 assign a unique set of four name servers (know as Collectively as a delegation set) to each hosted zone that you create

Example

Ns-1334.awsdns-39.com

Ns-895.awsdns-47.net

Ns-482.awsdns-53.org

Ns-1597.aws.dns-34.uk

Route 53 as your Authoritative DNS

- Once you update the route 53 ns (name server) settings with your domain register to include with route 53 name server route 53 will be responsible to respond to DNS Queries for the hosted zone
- This is true whether you do have a functionality website or not
- Route 53 will respond will information about the hosted zone whenever someone types the associated domain name in a web browser
- You can create more than 1 hosted zone with the same name and add different records to each hosted zone
- The name server are different for each of them
- When you update your registers name server record be careful to use the route 53 name server for the correct hosted zone the one that contains the records that you want that contains the records that you want route 53 to use when responding to queries for your domain
- Route 53 never returns values for records in another hosted zone that have the same name

Route 53 hosted zone by default you have two entries

- Ns entry – contains the unique sets of name server for this hosted zone
- SOA – contains information about the hosted zone

If you are currently using another DNS services and y

ou want to migrate to amazon Route53

- Start by creating a hosted zone
 - Route 53 automatically assigns the delegation sets the four-name server to your hosted zone
- To ensure that the DNS route queries for your domain to the route 53 name server
- Update your registers or your DNS services ns (name server records) domain to replace the current name servers with the name of the four route 53 name server for your hosted zone
- The method that you use to update the ns (name server records depend on which registrar or DNS service you are using)
- Some registrar only allows you to specify name server using Ip address they do not allow to specify fully qualified domain names
- If you registrar requires using Ip address you can get the Ip address for your name servers using the dig utility (for mac Linux) and nslookup (for window)

Transferring a domain between accounts within AWS

- Transferring a domain to a different aws account
 - If you registered a domain using one aws account and you want to transfer the domain to another aws account you can do so by contacting the aws support centre and requesting the transfer the domain

Migrating a hosted zone to different aws account

- If you are using route 53 as the DNS services for the domain route 53 does not transfer the hosted zone when you transfer a domain to a different aws account
- If domain registration is associated with one account and the corresponding hosted zone is associated with another account neither domain registration nor DNS functionality is affected
- The only effect is that you will need to sing into the route 53 console using one account to the domain and sign-in using the other account to see the hosted zone

Supported DNS records types by Route 53

1. A Records

- Address Record – maps domain name to Ip address
www.crackdevjob.com
- A -> 32 bits, ipv4

2. AAAA Records

- Ipv6 address records map domain name in to the ipv6 address

3. CNAME Records (canonical name record)

- Maps an alias to a hostname
- **Web in name** www.crackdevjob.com
- **Is an alternative records alias for another records**

4. Ns Records

- Name server records used for delegating zone to a name server

5. SOA (START OF Authority)

- Every single zone has one and only one SOA resources records at the beginning of the zone

It is not an actual record it includes the following information

- Who the owner is (email for the domain)
- The authoritative server
- The serial number which is incremental with changes to the zone data
- The refreshing time/cycle info, and the TTL (time to live)

6. MX Records

- Mail exchange defines where to deliver mail for user @ a domain name
- This is widely use for the mail

Routing Policies

When you create a Records, you choose a routing policies which determines how amazon route 53 responds to queries

➤ Routing Policies

- **Simple Routing (Default)**
- **Failover routing**
- **Geolocation routing**
- **Multi values answer routing**
- **Latency based routing**
- **Weighted routing**
- **Geo-proximity Routing**

➤ Failover Routing Policies

- **Failover routing** tells you route traffic to a resource when the resources is healthy if the main resources is not health, then route traffic to a different resource
- The Primary (active) and secondary (passive) records can route traffic to anything from an amazon s3 bucket that is configured as a website to a complex tree of records
- Failover routing policies is applicable for **public hosted zone only**

➤ Geolocation Routing Policies

- **Geo** location routing lets you choose the resources that serves your traffic based on the geographic location of your users i.e. the location that DNS queries originate from
- For e.g., you may have presence in Europe and Asia now you want users in the Aisa to be served in the Asia and those in Europe to be served by servers in Europe

Benefits

- You can localize your content and present some or all your website in the language of your users
- You can also use geolocation routing to restrict distribution of content to only the location in which you have distribution rights
- You can specify geographic location by continent by country or by state in the United States
- If you create sperate records for overlapping geographic regions for e.g. one records for north America and one for Canada – priority goes to the smallest geographic region
- Geolocation works by mapping Ip address to location however some Ip address are not mapped to geographic location

Latency Based Routing

- If your application is hosted in multiple amazon ec2 regions, you can improve performance for your users by serving their request from the amazon ec2 region that provide the lowest latency
- To use latency-based routing you create latency records for your resources in multiple ec2 regions

➤ When amazon route 53 receives a DNS query for your domain or subdomain

- It determines which amazon ec2 region you have created latency record for
- Determine which regions gives lowest latency to users
- Then select a latency record for that region

Weighted Routing Policy

- Weighted Routing Policy lets you associate multiple resources with single domain or subdomain and choose how much traffic is routed to each resource
- This can be useful for a variety of purpose including load balancing and testing new version of software
- Weight can be assigning any number from 1 to 255
- Weighted routing policy can be applied when there are multiple resources that perform the same function of e.g. Web server serving the same website
- To configure weighted Routing, you create Record that have the same name and type for each of your resources
- Amazon Route 53 send traffic to a resource based on the weight that you assigned to the records as a proportion in the group
- For e.g. suppose for www.crackdevjob.com has three resources record sets with weight, of 1(20 %), 1(20%) and 3 (60%) (sum = 5)
- On average Route 53 select each of the first two resource record set one fifth of the time and return the third resources records set three -fifth of the time
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